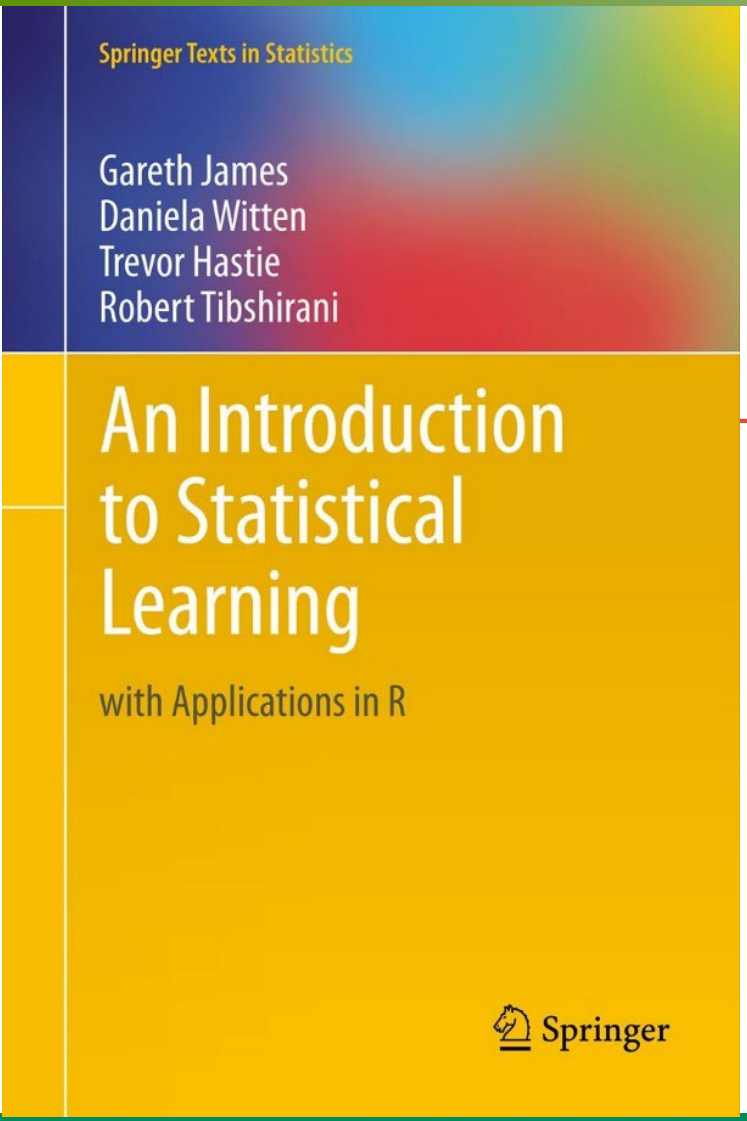


ANALYTICS

Week 3-4

Instructor:

Jeffrey Zheng

Examples Inspired By CAS, SOA, Coaching Actuaries ©

Key Formulas This Week

- Residual standard error: $s = RSE = \sqrt{MSE} = \sqrt{\frac{RSS}{n-2}}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$, $se(b_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$
- $se(\hat{y}) = s\sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$
- $se(\hat{y}_{pred}) = s\sqrt{1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{(n-1)s_x^2}} = s\sqrt{1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$
- Confidence Interval: $b_1 \pm t_{n-2,\frac{\alpha}{2}} * se(b_1)$
- Prediction Interval: $\hat{y}_{pred} \pm t_{n-2,\frac{\alpha}{2}} * se(\hat{y}_{pred})$

Hypothesis Testing

- Hypothesis testing helps us compare 2 models statistically.
- H_0: The simpler baseline model typically
- H_a: More complex model
- Goal: Is the additional complexity in H_a significant or "worth it" statistically
- How: Calculate a test statistic and compare to a threshold to see if we should reject H_0.

Hypothesis Testing 2

- For basic linear regression $Y = \beta_0 + \beta_1 X + \varepsilon$,

$\hat{y} = b_0 = \bar{y}$

- $H_0: \beta_1 = 0$ (no longer includes the x variable)

- H_a: $\beta_1 \neq 0$ $(\hat{y} = b_0 + b_1 x)$

t-statistic

- Use the t-statistic to help determine significance levels of the hypothesis test.
- t-ratio $=$ $\dfrac{\text{Estimator} - H_0 \text{ value}}{\text{se(Estimator)}}$
- For testing $H_0: \beta_1 = 0$ in basic linear regression,
- $t = \dfrac{b_1}{se(b_1)}$ with $df = n - 2$

Estimator Variance *

- The uncertainty around the estimated values $b_0, b_1, \hat{y}$ can be quantified using their respective standard errors.
- $Var(b_0) = MSE\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $Var(b_1) = MSE\left(\frac{1}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $Var(\hat{y}) = MSE\left(\frac{1}{n} + \frac{(x - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$ training
- $Var(\hat{y}_{pred}) = MSE\left(1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$ future

Standard Errors

- Standard error is the $\sqrt{\text{Variance}}$
- More observations (larger n), smaller residual standard deviation $s = \sqrt{MSE}$, and more leverage (higher s_x) will reduce se.
- $se(\hat{y}_{pred})$ is typically much larger due to the 2 sources of variability:
- 1. Uncertainty in estimating the regression line
- 2. Random noise error term ε

1. Var Illustration (SOA)

- Given a SLR model: $Y = \beta_0 + \beta_1 x + \varepsilon$ with 22 observations, the sample correlation coefficient is 0.5, and the estimated slope coefficient is 2.
- Determine the variance of the slope coefficient.

$$Var(b_1) = MSE\left(\frac{1}{\sum (x_i - \bar{x})^2}\right)$$

$$= \frac{SSE}{n-2}\left(\frac{1}{\sum (x_i - \bar{x})^2}\right) = \frac{SST(1-R^2)}{n-2}\left(\frac{b_1^2}{SSReg}\right)$$

$R^2 = .5^2 = 1 - \frac{SSE}{SST} \rightarrow SSE = SST(1-R^2)$

$= \frac{SSReg}{SST}$

$SSReg = b_1^2 \sum (x_i - \bar{x})^2$

$SS\,Reg = R^2(SST)$

$Var(b_1) = \frac{1-.5^2}{22-2}\left(\frac{2^2}{.5^2}\right) = 0.6$

1. Var Illustration 2

- $Var(b_1) = MSE\left(\frac{1}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $MSE = \frac{SSE}{n-2}$
- $R^2 = 1 - \frac{SSE}{SST} = \frac{SSReg}{SST}$
- $SSReg = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 = b_1^2 \sum_{i=1}^{n}(x_i - \bar{x})^2$

Example 1 (SOA)

- You use a simple linear regression model to predict Minecoins earned using # of hours played as a predictor, based on 9 observations. $n=9$
- $b_1 = \sqrt{20}$
- $\sum x_i = 27$
- $\sum x_i^2 = 91$
- $\sum (y_i - \hat{y}_i)^2 = 210$ ← SSE
- Calculate the t-statistic used for testing the hypothesis $H_0: \beta_1 = 1$.

$s^2 = MSE = \frac{SSE}{n-2} = \frac{210}{9-2} = 30$

$$\bar{x} = \frac{27}{9} = 3$$

$$se(b_1) = \frac{s}{\sqrt{\sum x_i^2 - n\bar{x}^2}} = \frac{\sqrt{30}}{\sqrt{91 - 9 \cdot 3^2}} = \sqrt{3}$$

$$t = \frac{b_1 - H_0}{se(b_1)} = \frac{\sqrt{20} - 1}{\sqrt{3}} = 2$$

Example 1 Sol

- $MSE = s^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2 = \frac{SSE}{n-2}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}} = \frac{s}{\sqrt{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}}$
- t-ratio $= \frac{Estimator - H_0\ value}{se(Estimator)}$

Var-Cov Matrix

- The variance-covariance matrix can be calculated to find standard errors.

- $Var(b) = \begin{bmatrix} Var(b_0) & Cov(b_0, b_1) \\ Cov(b_0, b_1) & Var(b_1) \end{bmatrix} =$

$MSE(\boldsymbol{X}^T\boldsymbol{X})^{-1}$

s^2

Degrees of Freedom

- Degrees of freedom reflects a model's flexibility and the # of unrestricted data points after using the model.
- In a simple linear regression with all predictor values known, if you have $n - 2$ points and the linear equation, the last 2 values are uniquely determined if the x-values are distinct.
- More flexible models have fewer df remaining

Test Statistic Decision

- Compare the calculated t to a threshold value (from t-Table) at the chosen significance α.
- Smaller α = (harder) Stricter criteria to reject H_0
- Reject H_0 if $|t| > t_{critical}$ for a 2-sided test, meaning $H_a: \beta_1 \neq 0$
- $t_{critical}$ comes from the t-Table

t-Table

- In SRM, the t-Table shows 1-sided α, so for 2-sided tests, double the Table's α.
- In MAS-I, the t-Table is 2-sided.

2-sided alpha	20%	10%	5%	2%	1%
df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$	$t_{0.010}$	$t_{0.005}$
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554

p-value

- The p-value is the probability of observing the data sample or a more extreme one by chance if H_0 were true

- If $p \leq \alpha$ or $|t| > t_{critical}$, we reject H_0.
- Rejecting H_0 does not mean our estimated b_1 is correct (typically never equal to the true), but rather that it adds value in the model

Hypothesis Testing Summary

- Set up H_0 & H_a. Decide on α.
- H_0 is a simpler model and H_a more complex.
- Use the estimated slope coefficient to determine your t statistic.
- Use the t-Table with $df = n - 2$ to determine critical t values at various α. Double the Table α for 2-sided tests.
- If the absolute value of the t statistic > critical t value at α, reject H_0 for that significance level and conclude parameter is useful to include.
- Smaller p-value or larger t = evidence to reject H_0

Test Relationships *

Larger (magnitude)

- Small $p \rightarrow$ Higher t-statistic, more evidence to reject $H_0 \rightarrow$ Keep parameter (it's significant)

- Large $p \rightarrow$ small t, fail to reject H_0, drop parameter

- The pre-set standard for rejecting H_0 is α.

F-statistic

- In basic linear regression, the F-statistic is used to test whether the single predictor is useful.
- $H_0: \beta_1 = 0$ vs. $H_a: \beta_1 \neq 0$
- $F = \frac{MSReg}{MSE} = (n-2)\frac{R^2}{1-R^2}$

- For basic linear regression, it is directly related to the t-statistic: $t = \sqrt{F}$

Poll Concept Check #1

- Which of the following statements are true about hypothesis testing under a simple linear regression?
- I. To find evidence for a negative slope parameter, the null hypothesis is that the slope parameter is less than 0.
- II. A two-tailed t-test will have the same degrees of freedom compared to a one-tailed t-test on the same dataset.
- III. If a t-test fails to reject the null hypothesis that the slope parameter is 0, then there is a relationship between the response and explanatory variables.
- IV. A large p-value leads to rejecting the null hypothesis.

2. t-test Illustration

- You fit the linear regression model: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ to 10 observations to model # of food allergies a person using $x_i = 1$ if the ith person's family member has a food allergy and $x_i = 0$ otherwise.
- 30% had a family member with a food allergy.
- $\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2 = 72$
- $\hat{\beta}_1 = 3$
- Calculate the t-statistic used to test the hypothesis: $H_0: \beta_1 = 0$ against $H_a: \beta_1 \neq 0$ and determine the smallest listed significance level α for which H_0 is rejected.

$$t = \frac{3-\phi}{se(b_1) \to 2.07} = 1.45 \qquad df = 10-2=8$$

$$se(b_1) = \frac{s}{\sqrt{\sum (x_i-\bar{x})^2}} = \frac{3}{\sqrt{2.1}} = 2.07$$

$$s = \sqrt{MSE} = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{72}{10-2}} = 3$$

7 $x_i = 0$, 3 $x_i = 7$

$$\sum (x_i-\bar{x})^2 = 7(0-.3)^2 + 3(1-.3)^2$$
$$= 2.1$$

$1.3968 < 1.45 < 1.8595$

reject at $\alpha = 20\%$ but

fail to reject at $\alpha = 10\%$

$.1 < p < .2$

2. t-test Illustration 2

- t-ratio $= \dfrac{Estimator - H_0\ value}{se(Estimator)}$
- $se(b_1) = \dfrac{s}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$
- $MSE = s^2 = \dfrac{1}{n-2}\sum_{i=1}^{n} e_i^2 = \dfrac{SSE}{n-2}$

2. t-test Illustration 3

- Table

2-sided alpha	20%	10%	5%	2%	1%
df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$	$t_{0.010}$	$t_{0.005}$
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554

Example 2 (CAS)

- A simple linear regression is run on 15 observations.
- $(X^T X)^{-1} = \begin{bmatrix} 0.5 & -0.2 \\ -0.2 & 0.1 \end{bmatrix} \cdot s^2$ → Var-Cov matrix
- $b_1 = 0.7$ ✓
- $\sum_{i=1}^{n}(y_i - \bar{y})^2 - b_1 \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = 20$ (SST − SSReg = SSE)
- Determine the smallest listed α where $H_0: \beta_1 = 0$ is rejected.

$$t = \frac{0.7 - 0}{se(b_1)} = \frac{0.7}{\sqrt{.15385}} = 1.785$$

df = 15 − 2 = 13

$$s^2 = MSE = \frac{SSE}{n-2} = \frac{20}{15-2} = 1.5385$$

$$Var(b_1) = .1s^2 = .15385$$

$1.7709 < 1.785 < 2.1604$

reject at $\alpha = 10\%$ but not $\alpha = 5\%$

Example 2 Sol

- $Var(b) = \begin{bmatrix} Var(b_0) & Cov(b_0, b_1) \\ Cov(b_0, b_1) & Var(b_1) \end{bmatrix} =$

  $MSE(\boldsymbol{X}^T\boldsymbol{X})^{-1}$

- $MSE = \frac{SSE}{n-2}$

Confidence Intervals

- A **confidence interval** gives a reasonable range that the true parameter value could fall within, given a significance level α.
- A $100(1-\alpha)\%$ confidence interval in basic linear regression is:
- $b_0 \pm t_{critical} * se(b_0)$
- $b_1 \pm t_{crit} \cdot se(b_1)$

Prediction Intervals

- A **prediction interval** gives a reasonable range for a particular prediction, given a significance level α.
- A $100(1-\alpha)\%$ prediction interval at x_{pred} in basic linear regression is:
- $\hat{y}_{pred} \pm t_{critical} * se(\hat{y}_{pred})$
- The prediction interval has to account for the uncertainty from the irreducible error ε that a confidence interval doesn't

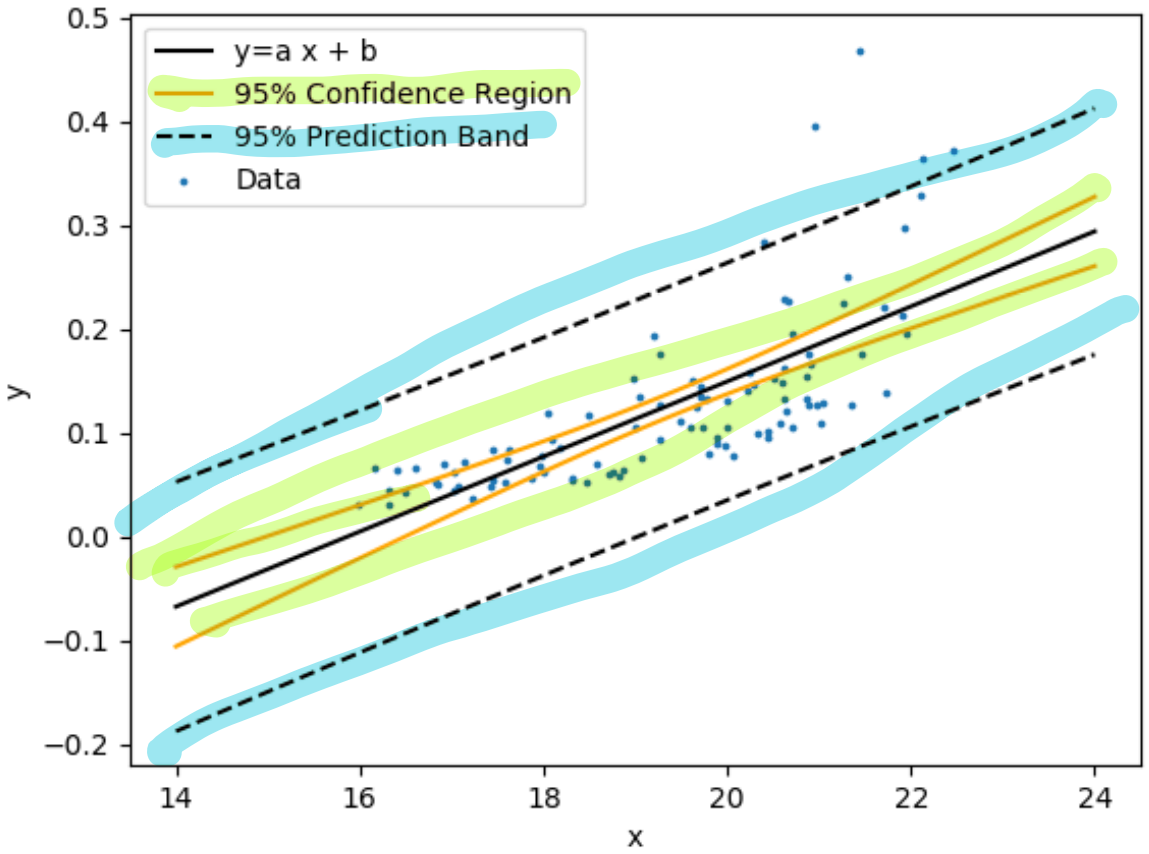
Prediction Assumptions

- Predictions assume the new observation follows the same data-generating model as the training sample

- Narrower confidence/prediction intervals are more informative.

Confidence vs. Prediction

- Both intervals are narrowest at $\bar{X}$.
- Prediction interval is at least as wide as confidence interval (due to irreducible error)



Intervals Summary

- Use confidence intervals for ____ $E[b_i|x]$ or $E[y|x]$
- Use prediction intervals for ____ $y|x$
- If a $100(1-\alpha)\%$ confidence interval for b_i includes 0, then we can conclude that ____ b_i is not significant at α, can be dropped from model (fail to reject $H_0: b_i = 0$)

Intervals and α

- The confidence interval and α represent two sides of the same coin.

Confidence Interval	80%	90%	95%	98%	99%
2-sided alpha	20%	10%	5%	2%	1%

df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$	$t_{0.010}$	$t_{0.005}$
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554

Poll Concept Check #2

- In our basic linear regression model, which statements are true?
- I. If there is no irreducible error, the confidence interval is equal to the prediction interval.
- II. The prediction interval quantifies the possible range of $E[b_1|x]$.
- III. The prediction interval is always at least as wide as the confidence interval.
- IV. The bias of ordinary least squares regression is 0.

3. Intervals Illustration

- A simple linear regression is fit on the following dataset to predict Y (in thousands) based on number of credentials X.
- Use your calculator as much as possible!
- A. Calculate the 95% confidence interval for β_1.
- B. Find the p-value range of the hypothesis test $H_0: \beta_1 = 4$.

$\bar{x} = 2.3$ $\quad \sum x^2 = 46, n = 6$

$b_0 = 52.375$

Y	50	60	65	70	75	80
X	0	1	3	2	4	4

$$b_1 \pm t_{crit} \cdot \frac{s}{\sqrt{\sum (x_i - \bar{x})^2}}$$

$t_{crit} = 2.7764$

"$\alpha = 6.125$"

$df = 6 - 2 = 4$

$\sum (x_i^2) - n\bar{x}^2$

$46 - 6 \cdot 2.\overline{3}^2$

$= 13.\overline{3}$

$$s = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{83.125}{6-2}}$$

$\hat{y}$'s (calculator):

$\hat{y}_i$	e_i^2
52.375	5.6406
58.5	2.25
70.75	33.063
64.625	28.891
76.875	3.5156
76.875	9.7656

$\sum x = 83.125$ (SSE)

95% CI b_1: $6.125 \pm 2.7764 \cdot \frac{\sqrt{20.78125}}{\sqrt{13.\overline{3}}}$

$\sqrt{13.\overline{3}} \approx 1.248$

$= (2.66, 9.59)$

3. Intervals Illustration 2

- $s^2 = \frac{SSE}{n-2}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}} = \frac{s}{\sqrt{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}}$
- $b_1 \pm t_{critical} * se(b_1)$
- t-ratio $= \frac{Estimator - H_0\ value}{se(Estimator)}$
- Do 2-Var Stats first.
- Data → Formula → Add/Edit Formula
- Predictions: L3 = b (from StatVars) + a*L1
- e^2: L1 (replace x) = (L2 – L3)^2 (will generate recursive error but still work)

(B) $t = \frac{6.125 - 4}{1.248} = 1.7$

reject H_0 at 20%

but not at 10%

3. Intervals Illustration 3

2 sided 20% 10%

df	$t_{0.100}$	$t_{0.050}$	$t_{0.025}$	$t_{0.010}$	$t_{0.005}$
1	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.8856	2.9200	4.3027	6.9646	9.9248
3	1.6377	2.3534	3.1824	4.5407	5.8409
4	1.5332	2.1318	2.7764	3.7469	4.6041
5	1.4759	2.0150	2.5706	3.3649	4.0321
6	1.4398	1.9432	2.4469	3.1427	3.7074
7	1.4149	1.8946	2.3646	2.9980	3.4995
8	1.3968	1.8595	2.3060	2.8965	3.3554
9	1.3830	1.8331	2.2622	2.8214	3.2498
10	1.3722	1.8125	2.2281	2.7638	3.1693
11	1.3634	1.7959	2.2010	2.7181	3.1058
12	1.3562	1.7823	2.1788	2.6810	3.0545
13	1.3502	1.7709	2.1604	2.6503	3.0123

Example 3 (SOA)

- You fit the model $y_t = \beta_0 + \beta_1 x_t + \epsilon_t$ to 10 observed values (x_t, y_t).
- $\sum(y_t - \hat{y}_t)^2 = 2.79$ SSE
- $\sum(x_t - \bar{x})^2 = 180$
- $\sum(y_t - \bar{y})^2 = 152.4$
- $\bar{x} = 6$
- $\bar{y} = 7.78$
- Determine the width of the shortest symmetric 95% prediction interval for y when $x = 8$.

Example 3 Sol

- Prediction Interval: $\hat{y}_{pred} \pm t_{critical} * se(pred)$
- $Var(\hat{y}_{pred}) = MSE\left(1 + \frac{1}{n} + \frac{(x_{pred} - \bar{x})^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right)$
- $MSE = s^2 = \frac{1}{n-2}\sum_{i=1}^{n} e_i^2 = \frac{SSE}{n-2}$

$df = 10 - 2 = 8$

$width = 2 \cdot t_{crit} \cdot se(pred) = 2.306 \cdot 0.6256 = 2.885$

$$s^2 = \frac{SSE}{10-2} = \frac{2.79}{8} = .34875$$

$$se(pred) = \sqrt{.34875} \cdot \sqrt{1 + \frac{1}{10} + \frac{(8-6)^2}{180}}$$

$$= .6256$$

Summary Check

- *Can you describe in your own words:*
- Formulas for standard error of $b_0, b_1, \hat{y}, \hat{y}_{pred}$
- How to conduct a t-test and use the table
- How to find confidence and prediction intervals